

1.Convert 2026, a base-10 number into binary, octal and hexadecimal system.

Given decimal number = 2026

1.Conversion into Binary (Base 2)

Divide the number by 2 repeatedly and note down the remainders.

$$2026 / 2 = 1013 \quad \text{remainder} = 0$$

$$1013 / 2 = 506 \quad \text{remainder} = 1$$

$$506 / 2 = 253 \quad \text{remainder} = 0$$

$$253 / 2 = 126 \quad \text{remainder} = 1$$

$$126 / 2 = 63 \quad \text{remainder} = 0$$

$$63 / 2 = 31 \quad \text{remainder} = 1$$

$$31 / 2 = 15 \quad \text{remainder} = 1$$

$$15 / 2 = 7 \quad \text{remainder} = 1$$

$$7 / 2 = 3 \quad \text{remainder} = 1$$

$$3 / 2 = 1 \quad \text{remainder} = 1$$

$$1 / 2 = 0 \quad \text{remainder} = 1$$

Reading the remainders from bottom to top,

$$2026_{10} = 11111101010_2$$

Therefore, the binary equivalent of 2026 is 11111101010.

(ii) Conversion into Octal (Base 8)

Divide the number by 8 repeatedly and note down the remainders.

$$2026 / 8 = 253 \quad \text{remainder} = 2$$

$$253 / 8 = 31 \quad \text{remainder} = 5$$

$$31 / 8 = 3 \quad \text{remainder} = 7$$

$$3 / 8 = 0 \quad \text{remainder} = 3$$

Reading the remainders from bottom to top,

$$2026_{10} = 3752_8$$

Therefore, the octal equivalent of 2026 is 3752.

(iii) Conversion into Hexadecimal (Base 16)

Divide the number by 16 repeatedly and note down the remainders.

$$2026 / 16 = 126 \quad \text{remainder} = 10 = A$$

$$126 / 16 = 7 \quad \text{remainder} = 14 = E$$

$$7 / 16 = 0 \quad \text{remainder} = 7$$

Reading the remainders from bottom to top,

$$2026_{10} = 7EA_{16}$$

Therefore, the hexadecimal equivalent of 2026 is 7EA.

Final Answer

Binary equivalent = 11111101010_2

Octal equivalent = 3752_8

Hexadecimal equivalent = $7EA_{16}$